

Homework 6: estimation

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Efficiency

The purpose of this exercise is to illustrate (empirically) that adopting different procedure of estimation you get different variability of the estimates.

1. Write a code in which you simulate 1000 samples of 40 observations from a Beta distribution with $\alpha = 2$ and $\beta = 0.7$ *Hint*. You can use the function `replicate`, see also lab 3.
2. Load the package `fitdistrplus` into R. The following function

```
estbeta.ml<-function(y)
{
  coefficients(fitdist(y,distr="beta"))
}
```

can be useful in the sequel. In fact calculate the estimates of α and β and plot the density histogram of the estimates.

4. Calculate the std.error of the distribution of the estimates.
5. The function `estbeta.ml` returns the ML estimates. Build a function `estbeta.mm` that returns the estimates by the method of the moments and calculate again the standard errors.
6. What is your conclusion ?

The TcCB Data

(Source: Millard, S.P. and Neerchal, N.K. (2001) *Environmental Statistics with S-PLUS*, CRC Press)

The guidance document *{Statistical Methods for Evaluating the Attainment of Cleanup Standards, Volume 3: Reference-Based Standards for Soils and Solid Media* (USEPA 1994b, pp. 6.22-6.25) contains 124 measures of 1,2,3,4- Tetrachlorobenzene (TcCB) concentrations (ppb) from soil samples at a Reference site and a Cleanup area (Table 1.1).

47 observations from the Reference site and 77 in the Cleanup area. These data are stored in the file `EPA.94b.tccb.df.txt`.

1. Import the dataset into R and create a dataframe named `EPA.94b.tccb.df`

In the dataset this observation is treated as censored at an assumed detection limit of 0.09 ppb (the smallest observed value).

There is one observation coded as “ND” in this data set as presented in the guidance document.

2. Select the TcCB concentrations for the Cleanup area
3. Consider a Box-Cox transformation of the selected TcCB concentrations such that the distribution of the transformed data looks similar to a Gaussian distribution

4. Fit a Gaussian distribution over the tranformed data.
5. Found the value (on the new scale) such that is overcome 1/100 times
6. Give the corresponding value on the ppb and name this a .
7. Suppose that you take three measures from the Cleanup Area. What is the probability to obtain at least one sample that overcome the value a ?

Tropical cyclones

(source: Elsner, J.B. and Jagger, T.H. (2012) *Hurricane Climatology*, Oxford University Press) The data set `US.txt` contains a list of tropical cyclone counts by year. The counts indicate the number of hurricanes hitting in the United States. The columns include year `Year`, number of U.S. hurricanes `All`, number of major U.S. hurricanes `MUS`, number of U.S. Gulf coast hurricanes `G`, number of Florida hurricanes `FL`, and number of East coast hurricanes `E`.

1. Consider the histogram of the counts for the U.S hurricanes and suggest a distribution that might be a good description for these data.
2. Estimate the parameter for this distribution.

Whales

(Source: Rice, J. (2007) *Mathematical Statistics and Data Analysis*, Duxbury Press)

The data of this exercise were gathered as part of a study to estimate the population size of the bowhead whale. The statistical procedures for estimating the population size along with an assessment of the variability of the estimate were quite involved, and this problem deals with only one aspect of the problem - a study of the distribution of whale swimming speeds.

Pairs of sightings and corresponding locations that could be reliably attributed to the same whale were collected, thus providing an estimate of velocity for each whale. The velocities, $v_1, v_2, \dots, v_{210}$ (km/h), were converted into times $t_1, t_2, \dots, t_{210}$ to swim 1 km . $t_i = 1/v_i$. The times are contained in the file `whales.txt`.

1. Plot the distribution of the t_i and
2. Does it appear that a gamma distribution would be a plausible model to fit?
3. Fit the parameters of the gamma distribution by the method of moments.
4. Fit the parameters of the gamma distribution by maximum likelihood. How do these values compare to those found before?
5. Plot the two gamma densities on top of the histogram. Do the fits look reasonable?